J. Symmetric Encoding

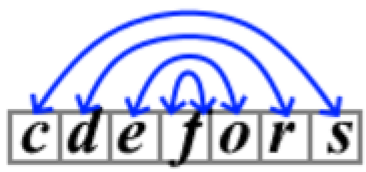
time limit per test: 2 seconds
memory limit per test: 256 megabytes

Polycarp has a string s, which consists of lowercase Latin letters. He encodes this string using the following algorithm:

- first, he constructs a new auxiliary string r, which consists of all distinct letters of the string s, written in alphabetical order;
- then the encoding happens as follows: each character in the string s is replaced by its symmetric character from the string r (the first character of the string r will be replaced by the last, the second by the second from the end, and so on).

For example, encoding the string s="codeforces" happens as follows:

- the string r is obtained as "cdefors";
- the first character s_1='c' is replaced by 's';
- the second character s_2='o' is replaced by 'e';
- the third character s_3='d' is replaced by 'r';
- ...
- the last character s_{10}='s' is replaced by 'c'.



The string r and replacements for s="codeforces".

Thus, the result of encoding the string s="codeforces" is the string "serofedsoc".

Write a program that performs decoding — that is, restores the original string s from the encoding result.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the length of the string b.

The second line of each test case contains a string b of length n, consisting of lowercase Latin letters — the result of encoding the original string s.

It is guaranteed that the sum of the values of n over all test cases in the test does not exceed $2 \cdot 10^5$.

Output

For each test case, output the string s from which the encoding result b was obtained.

Example

input Copy

```
5
10
serofedsoc
3
ttf
9
tlrhgmaoi
1
w
15
hnndledmnhlttin
```

output Copy

```
codeforces
fft
algorithm
w
meetinthemiddle
```